A Leave-one-specimen-out (piece never seen in training)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix	U-Net L1
A row7 col5 10x10 brain biological feature			LPIPS 0.832 PSNR 8.4 SSIM 0.35	LPIPS 0.732 PSNR 8.5 SSIM 0.19	LPIPS 0.743 PSNR 9.9 SSIM 0.03	LPIPS 0.695 PSNR 8.7 SSIM 0.23	LPIPS 0.930 PSNR 9.6 SSIM 0.37	LPIPS 0.807 PSNR 9.8 SSIM 0.37	LPIPS 1.133 PSNR 9.2 SSIM 0.37
D row6 col8 10x10 brain texture			LPIPS 0.616 PSNR 15.3 SSIM 0.35	LPIPS 0.679 PSNR 13.9 SSIM 0.21	LPIPS 0.796 PSNR 12.9 SSIM 0.41	LPIPS 0.824 PSNR 12.5 SSIM 0.38	LPIPS 0.765 PSNR 19.5 SSIM 0.53	LPIPS 0.591 PSNR 21.4 SSIM 0.50	LPIPS 0.852 PSNR 22.0 SSIM 0.56
E row6 col10 10x10 brain biological feature			LPIPS 0.649 PSNR 17.5 SSIM 0.58	LPIPS 0.629 PSNR 16.9 SSIM 0.42	LPIPS 0.646 PSNR 13.7 SSIM 0.45	LPIPS 0.635 PSNR 17.3 SSIM 0.50	LPIPS 0.720 PSNR 17.7 SSIM 0.54	LPIPS 0.629 PSNR 20.0 SSIM 0.58	LPIPS 0.779 PSNR 20.2 SSIM 0.60
G row3 col2 5x5 brain biological feature			LPIPS 0.621 PSNR 14.3 SSIM 0.32	LPIPS 0.676 PSNR 10.9 SSIM 0.27	LPIPS 0.840 PSNR 11.4 SSIM 0.34	LPIPS 0.889 PSNR 11.0 SSIM 0.37	LPIPS 0.776 PSNR 12.5 SSIM 0.40	LPIPS 0.680 PSNR 13.7 SSIM 0.41	LPIPS 0.844 PSNR 14.5 SSIM 0.43
H row4 col6 10x10 brain biological feature			LPIPS 0.694 PSNR 16.4 SSIM 0.46	LPIPS 0.556 PSNR 16.8 SSIM 0.27	LPIPS 0.607 PSNR 17.2 SSIM 0.19	LPIPS 0.506 PSNR 17.1 SSIM 0.45	LPIPS 0.743 PSNR 17.8 SSIM 0.55	LPIPS 0.567 PSNR 16.6 SSIM 0.45	LPIPS 0.753 PSNR 17.5 SSIM 0.53
I row3 col3 10x10 brain biological feature			LPIPS 0.518 PSNR 19.7 SSIM 0.50	LPIPS 0.461 PSNR 20.9 SSIM 0.59	LPIPS 0.723 PSNR 13.0 SSIM 0.43	LPIPS 0.393 PSNR 20.7 SSIM 0.63	LPIPS 0.843 PSNR 14.8 SSIM 0.36	LPIPS 0.776 PSNR 16.3 SSIM 0.37	LPIPS 0.710 PSNR 19.9 SSIM 0.58
B row6 col3 10x10 heart texture			LPIPS 0.804 PSNR 15.2 SSIM 0.38	LPIPS 0.570 PSNR 16.2 SSIM 0.31	LPIPS 0.651 PSNR 13.7 SSIM 0.34	LPIPS 0.460 PSNR 16.8 SSIM 0.37	LPIPS 0.952 PSNR 15.9 SSIM 0.40	LPIPS 0.643 PSNR 15.1 SSIM 0.39	LPIPS 0.830 PSNR 13.3 SSIM 0.38
C row1 col1 5x5 heart biological feature			LPIPS 0.550 PSNR 17.0 SSIM 0.42	LPIPS 0.591 PSNR 17.5 SSIM 0.30	LPIPS 0.761 PSNR 13.8 SSIM 0.34	LPIPS 0.767 PSNR 17.1 SSIM 0.43	LPIPS 0.787 PSNR 8.1 SSIM 0.34	LPIPS 0.663 PSNR 19.4 SSIM 0.44	LPIPS 0.732 PSNR 19.1 SSIM 0.47
F row4 col4 10x10 heart texture			LPIPS 0.508 PSNR 13.6 SSIM 0.28	LPIPS 0.502 PSNR 15.3 SSIM 0.16	LPIPS 0.802 PSNR 12.2 SSIM 0.37	LPIPS 0.657 PSNR 14.0 SSIM 0.40	LPIPS 0.768 PSNR 20.5 SSIM 0.47	LPIPS 0.619 PSNR 13.9 SSIM 0.37	LPIPS 0.660 PSNR 15.1 SSIM 0.38
J row0 col4 5x5 heart texture			LPIPS 0.587 PSNR 17.6 SSIM 0.45	LPIPS 0.488 PSNR 19.8 SSIM 0.40	LPIPS 0.869 PSNR 13.6 SSIM 0.38	LPIPS 0.695 PSNR 18.0 SSIM 0.42	LPIPS 0.882 PSNR 8.9 SSIM 0.34	LPIPS 0.610 PSNR 17.0 SSIM 0.51	LPIPS 0.723 PSNR 16.5 SSIM 0.52
K row1 col1 10x10 heart texture			LPIPS 0.488 PSNR 18.6 SSIM 0.51	LPIPS 0.488 PSNR 18.3 SSIM 0.46	LPIPS 0.820 PSNR 11.9 SSIM 0.38	LPIPS 0.588 PSNR 13.3 SSIM 0.34	LPIPS 0.543 PSNR 16.4 SSIM 0.45	LPIPS 0.543 PSNR 17.9 SSIM 0.46	LPIPS 0.596 PSNR 18.9 SSIM 0.53

B Spatial split (held-out region of a piece seen in training; within-specimen interpolation)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
A row7 col5 10x10 brain biological feature			LPIPS 0.771 PSNR 9.3 SSIM 0.35	LPIPS 0.535 PSNR 11.0 SSIM 0.29	LPIPS 0.529 PSNR 11.3 SSIM 0.27	LPIPS 0.511 PSNR 11.0 SSIM 0.34	LPIPS 0.836 PSNR 9.3 SSIM 0.27	LPIPS 0.609 PSNR 9.2 SSIM 0.30	LPIPS 0.619 PSNR 9.1 SSIM 0.29	LPIPS 0.868 PSNR 10.6 SSIM 0.37	LPIPS 0.712 PSNR 9.7 SSIM 0.37
D row6 col8 10x10 brain texture			LPIPS 0.579 PSNR 19.8 SSIM 0.45	LPIPS 0.560 PSNR 20.6 SSIM 0.46	LPIPS 0.560 PSNR 19.0 SSIM 0.44	LPIPS 0.530 PSNR 23.6 SSIM 0.52	LPIPS 0.621 PSNR 17.5 SSIM 0.30	LPIPS 0.804 PSNR 14.7 SSIM 0.38	LPIPS 0.783 PSNR 13.5 SSIM 0.21	LPIPS 0.749 PSNR 20.8 SSIM 0.55	LPIPS 0.567 PSNR 22.9 SSIM 0.52
E row6 col10 10x10 brain biological feature			LPIPS 0.572 PSNR 18.2 SSIM 0.55	LPIPS 0.530 PSNR 17.2 SSIM 0.51	LPIPS 0.530 PSNR 17.8 SSIM 0.53	LPIPS 0.508 PSNR 17.9 SSIM 0.52	LPIPS 0.649 PSNR 17.0 SSIM 0.46	LPIPS 0.592 PSNR 18.6 SSIM 0.34	LPIPS 0.576 PSNR 18.4 SSIM 0.54	LPIPS 0.905 PSNR 8.5 SSIM 0.34	LPIPS 0.569 PSNR 20.7 SSIM 0.56
G row3 col2 5x5 brain biological feature			LPIPS 0.544 PSNR 14.4 SSIM 0.32	LPIPS 0.611 PSNR 14.8 SSIM 0.36	LPIPS 0.594 PSNR 15.4 SSIM 0.34	LPIPS 0.602 PSNR 13.9 SSIM 0.39	LPIPS 0.567 PSNR 15.2 SSIM 0.26	LPIPS 0.795 PSNR 12.8 SSIM 0.35	LPIPS 0.802 PSNR 13.4 SSIM 0.40	LPIPS 1.137 PSNR 8.2 SSIM 0.34	LPIPS 0.633 PSNR 13.9 SSIM 0.40
H row4 col6 10x10 brain biological feature			LPIPS 0.619 PSNR 16.5 SSIM 0.43	LPIPS 0.507 PSNR 17.2 SSIM 0.42	LPIPS 0.510 PSNR 15.1 SSIM 0.36	LPIPS 0.493 PSNR 17.8 SSIM 0.47	LPIPS 0.533 PSNR 16.7 SSIM 0.34	LPIPS 0.618 PSNR 13.4 SSIM 0.39	LPIPS 0.535 PSNR 16.8 SSIM 0.35	LPIPS 0.782 PSNR 17.8 SSIM 0.57	LPIPS 0.551 PSNR 17.8 SSIM 0.50
I row3 col3 10x10 brain biological feature			LPIPS 0.450 PSNR 21.2 SSIM 0.62	LPIPS 0.367 PSNR 17.8 SSIM 0.60	LPIPS 0.498 PSNR 19.4 SSIM 0.46	LPIPS 0.355 PSNR 20.9 SSIM 0.63	LPIPS 0.456 PSNR 21.4 SSIM 0.52	LPIPS 0.708 PSNR 14.4 SSIM 0.42	LPIPS 0.683 PSNR 19.3 SSIM 0.50	LPIPS 0.872 PSNR 10.6 SSIM 0.34	LPIPS 0.784 PSNR 17.6 SSIM 0.42
B row6 col3 10x10 heart texture			LPIPS 0.492 PSNR 15.2 SSIM 0.29	LPIPS 0.477 PSNR 15.2 SSIM 0.27	LPIPS 0.440 PSNR 13.8 SSIM 0.23	LPIPS 0.412 PSNR 15.5 SSIM 0.35	LPIPS 0.520 PSNR 15.5 SSIM 0.26	LPIPS 0.664 PSNR 13.5 SSIM 0.32	LPIPS 0.469 PSNR 15.2 SSIM 0.22	LPIPS 0.774 PSNR 18.8 SSIM 0.41	LPIPS 0.626 PSNR 16.5 SSIM 0.38
C row1 col1 5x5 heart biological feature			LPIPS 0.550 PSNR 17.5 SSIM 0.38	LPIPS 0.558 PSNR 13.7 SSIM 0.39	LPIPS 0.588 PSNR 16.5 SSIM 0.43	LPIPS 0.516 PSNR 18.0 SSIM 0.42	LPIPS 0.566 PSNR 18.1 SSIM 0.35	LPIPS 0.782 PSNR 14.5 SSIM 0.38	LPIPS 0.798 PSNR 13.1 SSIM 0.42	LPIPS 0.852 PSNR 10.2 SSIM 0.34	LPIPS 0.700 PSNR 14.4 SSIM 0.43
F row4 col4 10x10 heart texture			LPIPS 0.469 PSNR 17.6 SSIM 0.38	LPIPS 0.486 PSNR 17.6 SSIM 0.37	LPIPS 0.474 PSNR 14.1 SSIM 0.32	LPIPS 0.468 PSNR 20.3 SSIM 0.42	LPIPS 0.605 PSNR 16.1 SSIM 0.35	LPIPS 0.788 PSNR 14.3 SSIM 0.35	LPIPS 0.640 PSNR 18.8 SSIM 0.40	LPIPS 0.820 PSNR 22.3 SSIM 0.48	LPIPS 0.534 PSNR 20.6 SSIM 0.41
J row0 col4 5x5 heart texture			LPIPS 0.597 PSNR 19.4 SSIM 0.47	LPIPS 0.431 PSNR 19.6 SSIM 0.46	LPIPS 0.442 PSNR 19.5 SSIM 0.52	LPIPS 0.529 PSNR 18.8 SSIM 0.50	LPIPS 0.563 PSNR 18.4 SSIM 0.43	LPIPS 0.876 PSNR 12.4 SSIM 0.38	LPIPS 0.762 PSNR 17.4 SSIM 0.51	LPIPS 0.946 PSNR 9.6 SSIM 0.34	LPIPS 0.744 PSNR 13.5 SSIM 0.48
K row1 col1 10x10 heart texture			LPIPS 0.558 PSNR 18.1 SSIM 0.51	LPIPS 0.450 PSNR 19.1 SSIM 0.60	LPIPS 0.435 PSNR 18.0 SSIM 0.47	LPIPS 0.484 PSNR 17.5 SSIM 0.60	LPIPS 0.503 PSNR 18.0 SSIM 0.40	LPIPS 0.769 PSNR 13.7 SSIM 0.44	LPIPS 0.555 PSNR 13.8 SSIM 0.41	LPIPS 0.891 PSNR 10.3 SSIM 0.34	LPIPS 0.624 PSNR 17.0 SSIM 0.50